SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**
Question Count: 5

Code of Conduct

M.AJAY / 192421021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ajay.M

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Total Marks Awarded

Signature of the Faculty

he concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Test Rubric

	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limite coverage
les	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
n	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentia concepts
	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult

21/07/2020 ASSESSMENT - 01 - CO1 M.AJAY

CLOUD COMPUTING IT

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[…]scribe the evolution from standalone computing to cloud-[…]sed systems. Summarize key technological transitions that [en]abled cloud computing.

Standalone computer worked independently with local storage.

Then networks connected computers to share data

The internet made online communication possible

Virtualization allowed multiple virtual machines on one server.

Finally, cloud computing enabled users to access data and applications over the internet

[…]pret the working of hypervisors in virtualizati[on] […]entiate between Type 1 and Type 2 hypervisors wi[th] […]ples.

[H]ypervisor is a software that creates and man[ages] [virtu]al machines.

Type 1 Hypervisor	Type 2 Hypervisor
Runs directly on hardware	Runs on host […]
Faster and more secure	Easier to insta[ll] […] use
Used in data centers	Used on pers[onal] computers
[Exa]mple : VMare ESXi, Hyper-V.	

		Example: Virtual Box, VMware Workstation

3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.

A: An API (Application Programming Interface) is a set of rules that allows different cloud applications and services to communicate with each other.

API help integrate cloud services by enabling the exchange of data and functions without requiring users to know the internal details of the system.

For example, a cloud storage service can use an API to connect with a cloud-based application and exchange files easily. This makes cloud systems more flexible, efficient, and easy to use.

Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.

* Resource pooling means sharing computing resources like servers and storage among many users

* Resources are allocated as needed.

* It reduces cost, improves efficiency, and supports many users at the same time.

*. supports multiple users at the same time without affecting performance.

Resource pooling is a cloud computing feature where computing resources such as servers, storage, memory, and networks are shared among multiple users.

6. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloa

A: Rapid Elasity means cloud resources can increase o decrease automatically based on demand.

<u>How it supports dynamic workloads:</u>

*. Resources can be scaled up during high demand
*. Resources can be scaled down when demand dec
*. Avoids resource wastage
*. Reduces operational wastage
*. provides flexibility to handle changing workloads.
*. Improves user experience by preventing service delay

Example: During an online shopping sale, cloud reso automatically increase to handle more users reduce after the sale ends.